

ALI NIKKHAH

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SUMMARY

Computer Vision Engineer with production experience spanning edge deployment, multimodal retrieval, and audio-visual intelligence systems. At Advanced Analytics Australia, delivered end-to-end surveillance pipelines on AXIS edge hardware, optimizing vision models with TensorRT and Triton and managing high-throughput streams via Apache Kafka. At Digikala, led multimodal product search using Vision Transformers and BLIP across millions of product assets. Research background includes achieving state-of-the-art results in automated medical report generation from ultrasound images (UBC) and texture-free video motion classification (L3S, UCF-101/HMDB-51).

TECHNICAL SKILLS

Languages & Runtimes: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
Data Engineering: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
MLOps & Serving: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
DevOps & Infrastructure: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
Databases & Monitoring: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

PROFESSIONAL EXPERIENCE

Advanced Analytics Australia

Remote Contract

Computer Vision Engineer

Sep 2025 – Apr 2026

- Engineered production video analytics pipelines on AXIS network camera hardware for health, safety, and compliance monitoring across **12 industrial sites**: handwash verification, PPE detection (boots, helmets), face tracking, and restricted-zone entry alerting (**50ms inference, 99.2% uptime**).
- Compiled and quantized vision models into **TensorRT** and **TensorFlow Lite** runtimes; served them through **Triton Inference Server** with dynamic batching, substantially reducing CPU overhead on edge nodes by eliminating Python-level inference overhead (**3.5x speedup vs Python**).
- Architected a high-throughput spatial metadata ingestion broker using **Apache Kafka** to buffer concurrent high-frame-rate tracking streams from **100+ cameras (30 FPS each)** without back-pressure on inference workers.
- Implemented dynamic sensor calibration routines that compensate for exposure variance and lighting shifts across indoor and outdoor camera placements, maintaining stable detection thresholds across all deployment sites (**-35% false positives**).
- Built asynchronous Python streaming workers interfacing directly with camera APIs, managing parallel video payload processing without blocking main inference threads.

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Led development of the multimodal product retrieval system, aligning image, diagram, and text product assets in a shared vector space using **Vision Transformers (ViT)** and **BLIP** encoders to enable cross-modal text-to-image semantic search across **10M+ product listings (+18% CTR, p99 150ms)**.
- Built high-throughput image preprocessing and spatial augmentation workers that fully decouple raw upload ingestion from model inference, preventing pipeline bottlenecks during traffic spikes.
- Profiled and optimized matrix operations in multimodal alignment architectures to eliminate out-of-memory (OOM) failures under heavy concurrent load.
- Supervised multi-tenant Kubernetes deployment isolating vision model serving nodes from language generation clusters, ensuring independent horizontal scaling and fault containment (**40% GPU cost reduction**).

Digikala

Tehran, Iran

AI Engineer

Nov 2024 – Jan 2025

- Prototyped and benchmarked multimodal indexing strategies for the product catalog; evaluated ViT versus CNN feature extractor throughput and cross-modal alignment quality to inform the architecture adopted in the lead role.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Scientist & AI Engineer

Sep 2025 – Mar 2026

- Engineered a telephony voice intelligence pipeline: live customer call audio is captured, transcribed in real time via **Whisper** and **wav2vec2.0**, and routed into a LangGraph agent for automated support ticket creation (**-78% manual tickets, 94% transcription accuracy**).
- Compiled acoustic feature extraction blocks to **TorchScript** and built Asyncio workers that process parallel voice payloads without blocking API threads, achieving low-latency transcription under simultaneous call load (**50+ concurrent streams, <100ms latency**).
- Designed continuous validation monitors that track signal-to-noise ratios on incoming audio streams, dynamically adjusting transcription thresholds to maintain accuracy under degraded telephony conditions.

HomaCloud

MLOps Engineer

Tehran, Iran

Aug 2021 – Feb 2022

- Orchestrated distributed model training and data pipelines on Kubernetes using Ray for scheduling and MLflow for experiment tracking; built low-latency model inference APIs with FastAPI and TensorFlow Serving.
- Automated full CI/CD lifecycle with Docker, ArgoCD, and GitHub Actions, cutting deployment cycle time to under 10 minutes; provisioned infrastructure-as-code with Terraform (**15+ models, 99.9% deploy success rate**).

SELECTED PROJECTS & RESEARCH

Automated Ultrasound Report Generation

ML Researcher

University of British Columbia

Feb 2024 – Nov 2025

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** trained with self-supervised contrastive learning to automatically generate DICOM-compliant radiology reports from raw ultrasound scans.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the benchmark; preprint in preparation.

Texture-Free Video Motion Classification

Research Assistant

L3S Research Center

Apr 2023 – Feb 2024

- Designed motion classification pipelines using 3D CNNs and transformer encoders over optical flow fields, deliberately suppressing texture cues to force learning of pure human-dynamic representations; achieved SOTA on **UCF-101** and **HMDB-51**.

RESEARCH HIGHLIGHTS

University of British Columbia

Research Assistant – Vision-Language Medical Imaging

Remote Research Collaboration

Feb 2024 – Nov 2025

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

L3S Research Center (Leibniz University Hannover)

Research Assistant – Video Motion Classification

Remote Research Collaboration

Apr 2023 – Feb 2024

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

EDUCATION

Sharif University of Technology

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Tehran, Iran

Sept 2020 – Jul 2024